Problem Set 5: Diffusion and Criticality

Due: Friday, February 20, 2026

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1. **Infinite Multiplication in a Graphite Pile (Lecture 12)**
   Consider a large, heterogeneous lattice of Natural Uranium fuel rods embedded in Graphite. You are given the following parameters for the lattice unit cell:

   - Fast Fission Factor: $\epsilon = 1.03$
   - Resonance Escape Probability: $p = 0.89$
   - Thermal Utilization: $f = 0.88$

   The reproduction factor η depends on the fuel. Natural Uranium is 0.72% ^{235}U and 99.28% ^{238}U.

   (a) Calculate the value of η for Natural Uranium. Assume that for pure ^{235}U, $\nu = 2.42$, $\sigma_f = 582$ b, and $\sigma_a = 680$ b. For ^{238}U, assume absorption is negligible at thermal energies relative to fission in ^{235}U (or use $\sigma_a^{238} \approx 2.7$ b if you prefer precision, but the simplified $\eta \approx \nu \frac{\Sigma_f^{235}}{\Sigma_a^{235}+\Sigma_a^{238}}$ is acceptable).

   (b) Calculate the Infinite Multiplication Factor k_∞.

   (c) Is this lattice capable of sustaining a chain reaction if the reactor is made infinitely large?

2. **Calculating Transport Properties (Lecture 13)**
   To solve the diffusion equation, we need the macroscopic transport properties D and L. Consider a homogeneous mixture of fuel and moderator with the following macroscopic cross-sections:

   - Transport (Scattering) Cross-section: $\Sigma_{tr} = 0.45$ cm^{-1}
   - Absorption Cross-section: $\Sigma_a = 0.005$ cm^{-1}

   (a) Calculate the **Transport Mean Free Path** λ_{tr} (in cm).

   (b) Calculate the **Diffusion Coefficient** D (in cm). *Recall the warning about units from Lecture 13!*

   (c) Calculate the **Diffusion Length** L (in cm). This represents the average "crow-flight" distance a neutron travels from birth to absorption.

3. **Geometric Buckling and Critical Dimensions (Lecture 14)**
   Using the material from Problem 2, assume the mixture has an infinite multiplication factor of $k_\infty = 1.05$.

   (a) Calculate the **Material Buckling** B_m^2 (in cm^{-2}).

   (b) We wish to build a **Spherical** reactor using this mix. Calculate the Critical Radius R_c (in meters).

   (c) We wish to build a **Cubic** reactor instead. Calculate the Critical Side Length a_c (in meters). Note: For a cube, $B_g^2 = 3(\pi/a)^2$.

   (d) Compare the **Total Volume** of the critical sphere vs. the critical cube. Which geometry is more efficient (requires less fuel)?

4. The Effect of a Reflector

A bare spherical reactor has a critical radius of $R_{bare} = 2.5$ m. We decide to surround the core with a thick Graphite reflector. The reflector modifies the boundary condition, effectively adding a "Reflector Savings" distance δ.

- Assume the Reflector Savings is $\delta = 25$ cm.

(a) What is the new physical radius $R_{reflected}$ of the fuel core required to maintain criticality?

(b) Calculate the percentage reduction in the **volume** of the expensive fuel core achieved by adding the reflector.

5. Criticality Safety in a Mixing Tank (Lecture 14)

You are designing a cylindrical mixing tank for a reprocessing plant (similar to the Cecil Kelley accident scenario). The tank has a fixed radius $R = 30$ cm. The solution being mixed has the following properties:

- $k_\infty = 1.25$
- Diffusion Length $L = 6.0$ cm

(a) Calculate the Material Buckling B_m^2.

(b) The Geometric Buckling for a finite cylinder is $B_g^2 = (\frac{2.405}{R})^2 + (\frac{\pi}{H})^2$. Calculate the "Radial Buckling" component $(\frac{2.405}{R})^2$.

(c) By equating $B_m^2 = B_g^2$, solve for the **Critical Height** H_c.

(d) If the tank is filled to a height of 50 cm, is the system Super-critical, Critical, or Sub-critical?

(e) **Safety Check:** If we increase the tank radius to $R = 50$ cm, what happens to the critical height? Does it increase or decrease? Explain why physically.